

Mutations induced by carcinogenic doses of aristolochic acid in kidney of Big Blue transgenic rats

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Abstract

Aristolochic acid (AA) is present in at least 65 different kinds of plants, many of which are used as herbal folk remedies. AA is considered one of the most potent plant carcinogens in humans and animals. It has been associated with the development of urothelial cancers in humans, and kidney and forestomach tumors in rats. In the present study, we used the Big Blue transgenic rat model to evaluate the mutagenicity of AA in kidney of rats and to define the mechanism of action for the tumor induction by AA. Groups of six male Big Blue transgenic rats were gavaged with 0, 0.1, 1.0 and 10.0 mg AA/kg body weight 5 times a week for 12 weeks, a treatment protocol that resulted in tumors in kidneys and other tissues. The animals were sacrificed 1 day after the final treatment and the kidneys were isolated for assays to determine the mutant frequencies (MFs) and types of mutations induced by AA in the transgenic *cII* gene. AA treatment resulted in a strong linear relationship between MF inductions and treatment dose ($R^2 = 0.998$). The *cII* MFs were $29 \pm 6 \times 10^{-6}$, $78 \pm 21 \times 10^{-6}$, $242 \pm 104 \times 10^{-6}$ and $1319 \pm 360 \times 10^{-6}$ in the control, low, medium and high dose treatment groups, respectively ($p < 0.001$ for all pair wise comparisons among the four treatment groups). These MFs correlated strongly with tumor incidences induced by the different doses of AA (Menges et al., 1982). Sequence analysis of the *cII* mutants revealed that there was a statistically significant difference between the mutational spectra in the AA-treated and control rats ($p < 0.05$). A:T $\rightarrow$ T:A transversion was the predominant type of mutation in the AA-treated rats whereas G:C $\rightarrow$ A:T transition was the main type of mutations in the control rats. These results suggest that AA induces kidney tumors in rats through a mutagenic mechanism of action.

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Keywords: Aristolochic acid; Mutagenicity; Kidney; Herb; Mutational spectrum; Carcinogenesis

1. Introduction

The Aristolochiaceae plant extract aristolochic acid (AA), a family of structurally related nitrophenanthrene

carboxylic acids, has been used as a herbal remedy by many cultures since antiquity. It is, however, among the most potent carcinogens of plant origin in animals and humans. The International Agency for Research on Cancer (IARC) has classified herbal remedies containing plant species of the genus *Aristolochia* as human carcinogens (IARC, 2002). AA is also among the most potent 2% of the carcinogens in Carcinogenic Potency and Genotoxicity Databases (Gold and Zeiger, 1997). AA

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is nephrotoxic and carcinogenic in humans and induces urothelial carcinoma among patients with Chinese-herb nephropathy (Cosyns et al., 1994; Nortier et al., 2000; Vanherweghem et al., 1995). Chinese-herb nephropathy is a progressive form of renal fibrosis associated with patients taking weight-reducing pills containing AA (Cosyns et al., 1994; Depierreux et al., 1994; Vanherweghem et al., 1993). AA also induces tumors in the kidney, forestomach, and other tissues of rats and mice (Mengs, 1983, 1988; Mengs et al., 1982).

The Food and Drug Administration (FDA) sent a letter to industry expressing its concern about botanical products containing AA, indicating that the use of these products resulted in several life-threatening adverse incidents in 2000 (FDA, 2000). FDA also advised consumers to immediately discontinue use of any botanical products containing AA and published a list of botanical products that contained AA in 2001 (Schwetz, 2001). Despite the actions of the FDA, many products containing or suspected to contain AA can still be identified on U.S. Web sites for sale for gastrointestinal symptoms, weight loss, cough and immune stimulation (Gold and Slone, 2003).

Although the mechanism of AA toxicity remains elusive, accumulating data suggest that reactive metabolites/intermediates formed via bioactivation of the major herbal constituents play a role in the herbal toxicity and carcinogenicity. The natural AA is a mixture of two major components, AAI and AAI. Metabolism studies demonstrate that the major metabolites of AAI and AAI in vitro and in vivo are the reduction products, aristolactam I and II, respectively (Krumbiegel et al., 1987; Schmeiser et al., 1986). These reactive metabolites can bind to DNA, and potentially induce mutations and initiate tumors (Arlt et al., 2002). AA induced structural chromosome aberrations and sister chromatid exchanges in human lymphocytes in vitro (Abel and Schimmer, 1983) and it was a direct-acting mutagen in *Salmonella typhimurium* strains (Pezzuto et al., 1988; Robisch et al., 1982; Schmeiser et al., 1984) and in L5178Y mouse lymphoma cells (Zhang et al., 2004). AA also induces mutations in forestomach, kidney, colon and bladder in mice (Kohara et al., 2002).

Although mutagenicity of AA has been widely studied in cells and mice, no study has been conducted on mutagenicity of AA in rats that have a strong response to AA for carcinogenesis (Mengs et al., 1982). In the absence of direct evidence on mutagenicity of AA in human, mutagenicity data from both rats and mice can provide better information for predicting the underlying mechanisms for mode of action of AA carcinogenesis because the confidence in extrapolation from rodents to

human can be strengthened if there is a good agreement between mouse and rat for the mutagenic responses.

In this study, we used Big Blue transgenic rat model to evaluate the mutagenicity of AA in the kidney *cII* gene and to define the mechanism of action for the tumor induction by AA. Big Blue transgenic rats provide a valuable tool for analyzing in vivo somatic cell mutation and permit a direct comparison of tumor incidence with the frequency and types of mutations induced by AA. We treated the animals with a similar protocol that resulted in tumors in rats for this mutagenesis study (Mengs et al., 1982) and measured the mutant frequency (MF) in the transgenic *cII* gene. We found that the MF induced by the different doses of AA in the kidneys of treated rats correlated with the tumor incidence. In addition, the mutational spectrum induced by AA in the kidney reflected the mutational specificity of AA-DNA adducts and the type of mutations found in the *H-ras* gene of AA-induced tumors (Schmeiser et al., 1990).

2. Materials and methods

2.1. Test agent and animals

Aristolochic acid (AA) was purchased from Sigma (St. Louis, MO). The purity of AA was 96% and included 40% of AAI and 56% of AAI. Big Blue transgenic Fisher 344 rats were obtained from Taconic Laboratories (Germantown, NY) through a purchase from Stratagene (La Jolla, CA). We followed the recommendations of the NCTR Institutional Animal Care and Use Committee for the handling, maintenance, treatment and sacrifice of the rats.

2.2. Treatments

The treatment schedule was based on the previous carcinogenicity study (Mengs et al., 1982). Groups of six 6 week-old male Big Blue rats were treated with AA as its sodium salt at concentrations of 0.1, 1.0 and 10.0 mg/kg body weight by gavage (4 ml/kg body weight) 5 times a week for 12 weeks. Vehicle control rats were gavaged with 0.9% sodium chloride using the same schedule. The rats were sacrificed 1 day after the last treatment; the kidneys were isolated, frozen quickly in liquid nitrogen and stored at -80°C .

2.3. *cII* Mutation assay

High molecular weight genomic DNA was extracted from rat kidneys using the RecoverEase DNA Isolation Kit (Stratagene) and stored at 4°C until DNA packaging was performed. The packaging of the phage, plating the packaged DNA samples, and determination of MF were carried out following the manufacturer's instructions for the Select-*cII* Mutation Detection System for Big Blue Rodents (Stratagene). The shuttle vector containing the *cII* target gene was rescued from

total genomic DNA with phage packaging extract (Transpack; Stratagene). The plating was performed with the *Escherichia coli* host strain G1250. To determine the total titer of packaged phages, G1250 bacteria were mixed with 1:3000 dilutions of phage, plated on TB1 plates, and incubated overnight at 37 °C (nonselective conditions). For mutant selection, the packaged phages were mixed with G1250, plated on TB1 plates, and incubated at 24 °C for about 42 h (conditions for *cII* selection). Under these conditions, phages with wild-type *cII* genes undergo lysogenization and become part of the developing bacterial lawn, whereas phages with mutated *cII* genes undergo lytic growth and give rise to plaques. When incubated at 37 °C, phages with both wild-type and mutated *cII* genes undergo a lytic cycle, resulting in plaque formation. Assays were repeated until a minimum of 2×10^5 plaque-forming units (pfus) from each sample was examined for mutation. The *cII* MF is defined as the total number of mutant plaques (determined at 24 °C) divided by the total number of plaques screened (determined at 37 °C).

2.4. Sequence analysis of the *cII* mutants

cII Mutant plaques were selected at random from different animals and replated at low density to verify the mutant phenotype. Single, well-isolated plaques were selected from these plates and transferred to a microcentrifuge tube containing 100 μ l of sterile distilled water. The tube was heated at 100 °C for 5 min and centrifuged at $12,000 \times g$ for 3 min. The *cII* target DNA for sequencing was amplified by PCR using primers 5'-AAAAAGGGCATCAAATTAACC-3' (upstream) and 5'-CCGAAGTTGAGTATTTTGCTG-3' (downstream). For PCR amplification, 10 μ l of the supernatant were added to 10 μ l of a PCR Master Mix (Promega, Madison, WI) and the primers. The final concentrations of the reagents were 1 $\times$ Taq polymerase reaction buffer, 0.2 μ mol of each primer, 200 μ M of each dNTP, 1.5 mM MgCl₂, and 0.25 U of Taq DNA polymerase. The PCR reaction was performed using a PCR System 9700 (Applied Biosystems, Foster City, CA), with the following cycling parameters: a 3 min denaturation at 95 °C; followed by 35 cycles of 30 s at 95 °C, 1 min at 60 °C, and 1 min at 72 °C; with a final extension of 10 min at 72 °C. The PCR products were isolated using a PCR purification kit (Qiagen, Chatsworth, CA). The *cII* mutant DNA was sequenced with a CEQ Dye Terminator Cycle Sequencing Kit and a CEQ 8000 Genetic Analysis System (Beckman Coulter, Fullerton, CA). The primers for *cII* mutation sequencing were the primers used for the PCR.

2.5. Statistical analyses

Analyses were performed using SigmaStat 2.03 (SPSS, Chicago, IL). All of the MF data are expressed as the mean $\pm$ standard deviation (S.D.) from six rats per group. Statistical significance was determined by one-way analysis of variance (ANOVA) followed by the Tukey test. Since the variance increased with the magnitude of the MF, the data were

log-transformed before conducting the analysis. Mutational spectra were compared using the computer program written by Cariello et al. (1994) for the Monte Carlo analysis developed by Adams and Skopek (1987).

3. Results

3.1. MFs induced by AA in kidney of rats

MFs in the kidney *cII* gene are shown in Fig. 1. The MFs of the control rats were between 19×10^{-6} and 35×10^{-6} , with an average of $29 \pm 6 \times 10^{-6}$. The MFs in the low, medium and high dose treatment groups were $78 \pm 21 \times 10^{-6}$, $242 \pm 104 \times 10^{-6}$ and $1319 \pm 360 \times 10^{-6}$, respectively, which were significantly higher than the control ($p < 0.001$) with an increase of about 3-, 8-, and 45-fold over the control. There was a strong linear dose response ($R^2 = 0.998$) and a statistically significant difference between any two of the three treatment groups ($p < 0.001$).

3.2. Mutational types induced by AA in kidney of rats

The *cII* genes from 92 mutants from the 6 high dose-treated rats and 48 mutants from the 6 control rats were examined for mutations (Table 1 and summarized in Table 2). Mutations that were found more than once in the mutants isolated from a single animal were assumed to be siblings and were considered to represent a single independent in vivo mutational event. Accordingly, 84 independent mutations from AA treatment and 41 from control were identified. Statistical

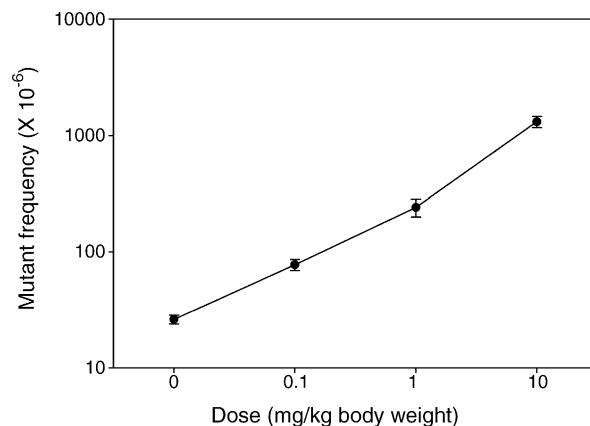


Fig. 1. Mutant frequencies induced by different doses of AA in the kidney *cII* gene of the Big Blue transgenic rat. Significant differences were found for all pair wise comparisons among the four different dose groups ($p < 0.001$, $n = 6$). $R = 0.994$ and $p = 0.002$ for the linear regression.

Table 1

Mutations in the kidney *cII* gene of Big Blue rats treated with aristolochic acid and vehicle control

Position ^a	Mutation ^b	Amino acid change	Sequence context 5' → 3' ^c	Number of mutations (independent)	
				Control	Treated
–14	G → T	N/A	ctaaggaaa	1	
–1	T → C	N/A	ttacat ^u atg		1
1	A → G	Met → Val	catATGg ^u tt		8 (6)
1	A → T	Met → Leu	catATGg ^u tt		3 (2)
2	T → C	Met → Thr	catATGg ^u tt	1	1
2	T → G	Met → Arg	catATGg ^u tt		1
3	G → A	Met → Ile	catATGg ^u tt		1
3	G → C	Met → Ile	catATGg ^u tt	1	
6/7	+G	Frameshift	atgGTTCTG ^u Tgca	1	
16	A → T	Lys → Stop	aacAAAcgc		1
25	G → A	Glu → Lys	aacGAGg ^u ct	2 (2)	1
28	G → A	Ala → Thr	gagGCTc ^u ta		1
28	G → C	Ala → Pro	gagGCTc ^u ta	1	1
29	C → A	Ala → Asp	gagGCTc ^u ta	1	
31	C → G	Leu → Val	gctCTAcga		1
37	A → G	Ile → Val	cgaATCgag		1
38	T → A	Ile → Asn	cgaATCgag		1
40	G → A	Glu → Lys	atcGAGag ^u t	2 (2)	
40	G → T	Glu → Stop	atcGAGag ^u t	1	
42	G → C	Glu → Asp	atcGAGag ^u t		1
50	T → A	Leu → Stop	gcgTTGc ^u tt		2 (2)
53	T → A	Leu → His	ttgCTTaa ^u c		1
56	A → T	Asn → Ile	cttAACa ^u aa		2 (2)
58	A → T	Lys → Stop	aacAAAtc		2 (2)
61	A → T	Ile → Phe	aaaATCgca		1
71	T → A	Leu → His	atgCTTgga		1
72	T → C	Leu → Leu	atgCTTgga		2 (2)
73	G → A	Gly → Arg	cttGGAact		2 (2)
74	G → A	Gly → Glu	cttGGAact	1	
83	A → T	Lys → Met	gagAAGaca		1
85	A → T	Thr → Ser	aagACAgcg		1
88	G → T	Ala → Ser	acaGCGgaa	1	
89	C → T	Ala → Val	acaGCGgaa	3 (2)	
94	G → T	Ala → Ser	gaaGCTgtg	1	
98	T → A	Val → Glu	gctGTGg ^u gc		3 (3)
100	G → A	Gly → Ser	gtgGCGgtt		1
103	G → C	Val → Leu	ggcGTTgat		2 (2)
103	G → T	Val → Phe	ggcGTTgat		1
103/104	+C	Frameshift	ggcGTTgat		1
104	T → G	Val → Gly	ggcGTTgat		1
108	T → A	Asp → Glu	gttGATaa ^u g		2 (2)
109	A → T	Lys → Stop	gatAAGtcg		2 (2)
110	A → T	Lys → Met	gatAAGtcg		2 (1)
111/112	+G	Frameshift	gatAAGTCGcag	1	
113	C → T	Ser → Leu	aagTCGcag	4 (1)	
113	C → G	Ser → Trp	aagTCGcag	2 (2)	
116	A → T	Gln → Leu	tcgCAGatc	1	4 (2)
125	G → A	Arg → Lys	agcAGGtgg	1	
126–127	GT → CA	ArgTrp → SerArg	agcAGGTG ^u Gaag	1	
127	T → A	Trp → Arg	aggTGGaag		1
128	G → A	Trp → Stop	aggTGGaag	1	1
130	A → G	Lys → Glu	tggAAGagg		1
131	A → T	Lys → Met	tggAAGagg		1
133	A → T	Arg → Trp	aagAGGgac		2 (2)
139	T → A	Trp → Arg	gacTGGatt		2 (2)

Table 1 (Continued)

Position ^a	Mutation ^b	Amino acid change	Sequence context 5' → 3' ^c	Number of mutations (independent)	
				Control	Treated
141	G → C	Trp → Cys	gacTGGatt	1	
141	G → T	Trp → Cys	gacTGGatt	1	
148	A → C	Lys → Gln	ccaAAGttc		1
149	A → T	Lys → Met	ccaAAGttc		1
152	T → A	Phe → Tyr	aagTTCta		1
152	T → G	Phe → Cys	aagTTCta		1
154	T → A	Ser → Thr	ttcTCAatg		2 (1)
157/158	+TG	Frameshift	tcaATGctg	1	
161	T → A	Leu → Gln	atgCTGctt		3 (3)
161	T → G	Leu → Arg	atgCTGctt	1	
163	C → T	Leu → Phe	ctgCTTgct	1	
164	T → A	Leu → His	ctgCTTgct		2 (1)
176	A → T	Glu → Val	cttGAAtgg		1
178/185	+G	Frameshift	gaaTGGGGGGTCgtt	8 (5)	
178	T → C	Trp → Arg	gaaTGGggg		1
179	G → A	Trp → Stop	gaaTGGggg		3 (3)
179	G → C	Trp → Ser	gaaTGGggg	1	
187	G → A	Val → Ile	gtcGTTgac		1
193	G → A	Asp → Asn	gacGACgac		1
193	G → T	Asp → Tyr	gacGACgac	1	1
200	T → C	Met → Thr	gacATGgct		1
200	T → A	Met → Lys	gacATGgct		1
209	T → A	Leu → Stop	cgaTTGgcg		1
210	G → T	Leu → Phe	cgaTTGgcg	1	1
212	C → A	Ala → Glu	ttgGCGcga	2 (2)	
212	C → T	Ala → Val	ttgGCGcga	2 (2)	
214	C → T	Arg → Stop	gcgCGAcaa		1
215	G → C	Arg → Pro	gcgCGAcaa		1
230	T → A	Ile → Asn	gcgATTctc		1
234–235	–CA	Frameshift	attCTCACCatt		1
Total				48 (41)	92 (84)

Abbreviations: –, deletion; +, insertion.

^a Position 1 is the first base of the start codon in the *cII* coding sequence.

^b Presented in term of sequence change on nontranscribed DNA strand.

^c Uppercase indicates target codon and target bases are underlined.

analysis revealed that there was a statistically significant difference between the mutation spectra in the AA-treated and control rats ($p < 0.00001$). A:T → T:A transversion (50% in AA treatment versus 2% in control) was the predominant mutation type for AA-treated rats, whereas the main types of mutations in control rats were G:C → A:T transitions (32% in control versus 15% in AA treatment), most of which occurred at CpG sites (69% of G:C → A:T transitions in control and 22% in AA treatment), and frameshifts (20% in control versus 1% in AA treatment).

4. Discussion

In order to compare the mutagenicity of AA with its carcinogenicity, we treated Big Blue rats with AA

using a protocol that results in tumors. The treatment resulted in a significant increase in MF and a strong linear dose response (Fig. 1). The general nature of the dose–response for MF was consistent with previous carcinogenesis study (Menges et al., 1982). We did not score the tumors because the tumors were not able to be identified at the time when we measured the MFs. Oral treatment with AA resulted in a dose- and time-dependency of induction of tumors. Incidence of tumors in male rats after 3 month treatment with 0, 0.1, 1.0 or 10.0 mg/kg/day AA was 0, 27, 86 and 100%, respectively, with the main target tissues forestomach, kidney, renal pelvis, and urinary bladder. The correlation between induction of mutations and tumors by AA is consistent with AA acting as a potent genotoxic carcinogen.

Table 2

Mutational spectra in the kidney *cII* gene of Big Blue transgenic rats treated with aristolochic acid and vehicle control

Type of Mutation	Control			Aristolochic acid		
	Number	%	MF ($\times 10^{-6}$)	Number	%	MF ($\times 10^{-6}$)
Transitions						
G:C $\rightarrow$ A:T	13	32	9.3	13	15	197.9
A:T $\rightarrow$ G:C	1	2	0.6	14	17	224.2
Transversions						
G:C $\rightarrow$ T:A	10	24	7.0	3	4	52.8
G:C $\rightarrow$ C:G	6	15	4.4	6	7	92.3
A:T $\rightarrow$ T:A	1	2	0.6	42	50	659.5
A:T $\rightarrow$ C:G	1	2	0.6	4	5	66.0
Frameshifts	8	20	5.8	1	1	13.2
Tandem base substitution	1	2	0.6	0	0	0
Total mutations	41	100	29	84	100	1319

Note: MF, mutant frequency.

Sequence analysis of the *cII* mutants from the high-dose treatment revealed that A:T $\rightarrow$ T:A transversion was the predominant mutation (Table 2). If grouping into mutations occurring at A:T sites, a larger difference can be observed between the treated and control groups, 73% versus 6%. This type of mutation is consistent with the specificity of the DNA adducts formed by AA in rat kidney. In our previous study, we identified three major AA DNA adducts, 7-(deoxyadenosin- N^6 -yl)aristolactam I (dA-AAI), 7-(deoxyguanosin- N^2 -yl)aristolactam I (dG-AAI) and 7-(deoxyadenosin- N^6 -yl)aristolactam II (dA-AAII) in rat kidney (Mei et al., 2005); these same adducts are also found in human urothelial tumors (Arlt et al., 2002, 2004). It was reported that dA-AAI adducts were very persistent while dG-AAI adducts were rapidly removed from rat DNA (Fernando et al., 1993). Also, dAMP and dTMP were preferentially incorporated into the places opposite the adenine adducts while dCMP was preferentially incorporated into the sites opposite guanine adducts, resulting A:T $\rightarrow$ T:A transversion, and A:T $\rightarrow$ A:T and G:C $\rightarrow$ G:C nonmutagenic events (Broschard et al., 1994, 1995). Therefore, the A:T $\rightarrow$ T:A transversions detected in rat kidneys most likely result from the dA-AA adducts that are persistent and able of leading to mutations.

Kohara et al. (2002) analysed the MF and mutational spectra in the *lacZ* and *cII* genes of Muta mice treated with 15 mg AA/kg body weight once a week for 4 weeks. They found that MF was increased 9-fold over the control by AA in kidney and the major mutations induced by AA was A:T $\rightarrow$ T:A transversions. Their data were consistent with our results. For example, in our medium dose treatment group (the total dose used was 60 mg/kg, the same as the total dose for the mouse treatment), AA

induced a 8-fold MF over the control. This consistence further supports the hypothesis that AA produces tumor via a mutagenic mechanism.

Although the *cII* gene in Big Blue rats is a transgene from λ virus, the mutations occur at this gene appear to be indicative of mutations that occur at endogenous genes like oncogenes. AA activates *H-ras* genes in rat tumors via A:T $\rightarrow$ T:A transversion mutation at the adenine in the middle of codon 61 (CAA); this particular mutation was detected in tumors induced by AA in rodents (Schmeiser et al., 1990, 1991). Activation of *H-ras* gene is associated with initiation of carcinogenesis. The correlation between mutations induced by AA in the *cII* gene and the *H-ras* gene, as well as between the tumor incidence and the *cII* MF induced by AA, suggests that the transgenic rat mutation assay can serve as a short term assay to predict the carcinogenicity of genotoxins.

In summary, AA is a potent mutagen in rat kidney and the AA-induced MFs correlate well with the incidence of tumors produced by AA in rats. The AA-induced mutations in rat kidney reflect the specificity of AA-DNA adducts and the predominant type of mutation is the same type of mutation that results in activation of *H-ras* and initiation of tumors. These results support a mutagenic mechanism of action for tumor induction by AA.

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